

Vehicle Localization & Tracking in a Highway Tunnel

Localization, Navigation and Smart Mobility
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Scenario: one base station, two arrays, a vehicle at 98 km/h



Vehicle 110 m sinuous path, 27 m/s (98 km/h), constant height $z = 1.5$ m



Base station Ceiling of the tunnel at $[0, 50, 4.8]$ m — halfway down the track



Two arrays Boresights $(az, el) = (-90^\circ, -30^\circ)$ and $(+90^\circ, -30^\circ)$; handover at $t = 59 \rightarrow 60$



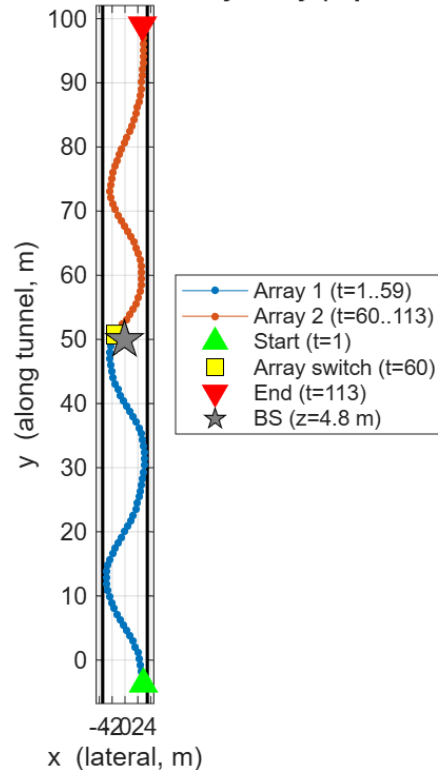
Sampling 113 steps at $T_s = 36$ ms; every step delivers 1–5 multipath rays



Per ray (range, azimuth, elevation) — true values for scoring, measured values for estimation

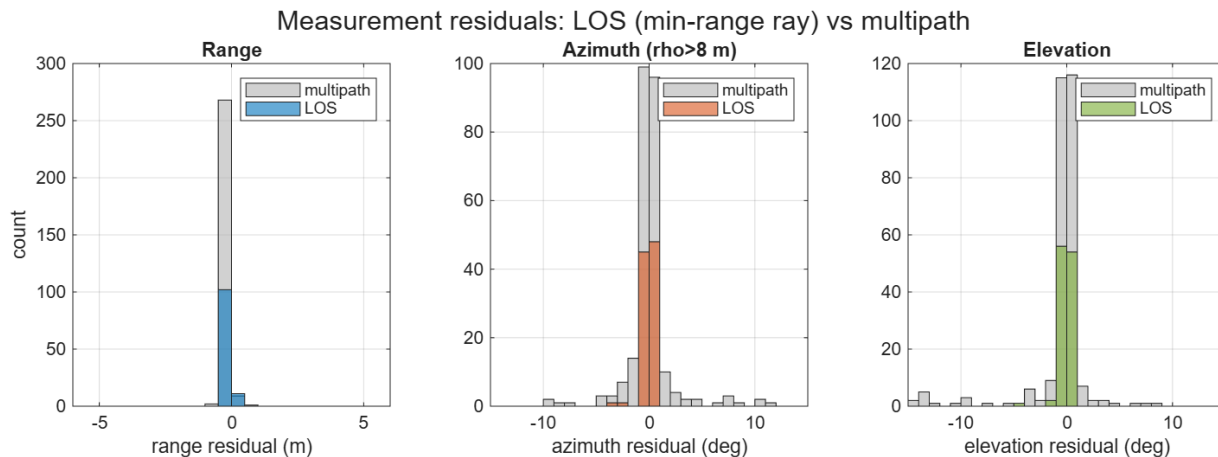
Geometry drives everything: range $\approx |y - 50|$ observes the along-tunnel axis; angles observe the lateral axis with lever arm $r \cdot 1^\circ \approx 0.9$ m at 50 m — strongest near the BS.

Tunnel scenario — true vehicle trajectory (top-down)





Task 1 — Data analysis: how accurate are the measurements?



Measured – true residuals after greedy NN ray association (LOS = min-range ray)

0.08 m

range σ (LOS)

0.46°

azimuth σ (far field)

0.49°

elevation σ (LOS)

Noise model → estimator R. Sigmas estimated from LOS residuals, floored to (0.20 m, 1°, 1°) for positioning — the min-range ray is only an approximate LOS (≤ 0.41 m long) and residual multipath leaks in. Model verified: predicted LOS matches truth to < 0.5 m / $< 1^\circ$ far-field; analytic Jacobian = finite differences to $1e-9$.



The two required issues: association & measurement selection



1 · Ray association (analysis)

- Ray counts differ between true and measured sets (1–5 per step) — no index correspondence.
- Greedy nearest-neighbour on normalized ($\Delta r/0.5$ m, $\Delta \text{az}/3^\circ$, $\Delta \text{el}/3^\circ$) squared distance.
- Used in Task 1 to characterize accuracy per ray.



2 · Measurement selection (positioning)

- NLS (memoryless): pick the minimum-range measured ray as the LOS.
- EKF: pick the Mahalanobis-nearest ray w.r.t. the prediction; reject if $d^2 > \chi^2_{0.99}$ (gate) → coast on the motion model.
- Angle innovations wrapped to $(-180^\circ, 180^\circ]$.



Why it matters — the ghost ray at $t = 19$. One measured ray claims range 6.38 m while the true distance is 37.47 m, with an impossible $+6^\circ$ elevation (real rays always arrive from above). Min-range selection swallows it → every NLS variant peaks at ~ 30 m error there. The EKF gate rejects it and coasts — error stays sub-metre.



Task 2 — Positioning: snapshot NLS vs EKF, three modes

NLS — snapshot Gauss-Newton

- Per step, no memory; prior-regularized ($\sigma = 50$ m) around the previous fix.
- Mode-honest init: ToA $\rightarrow$ range + known array side; AoA $\rightarrow$ angles + known height.

EKF — predict · gate · update

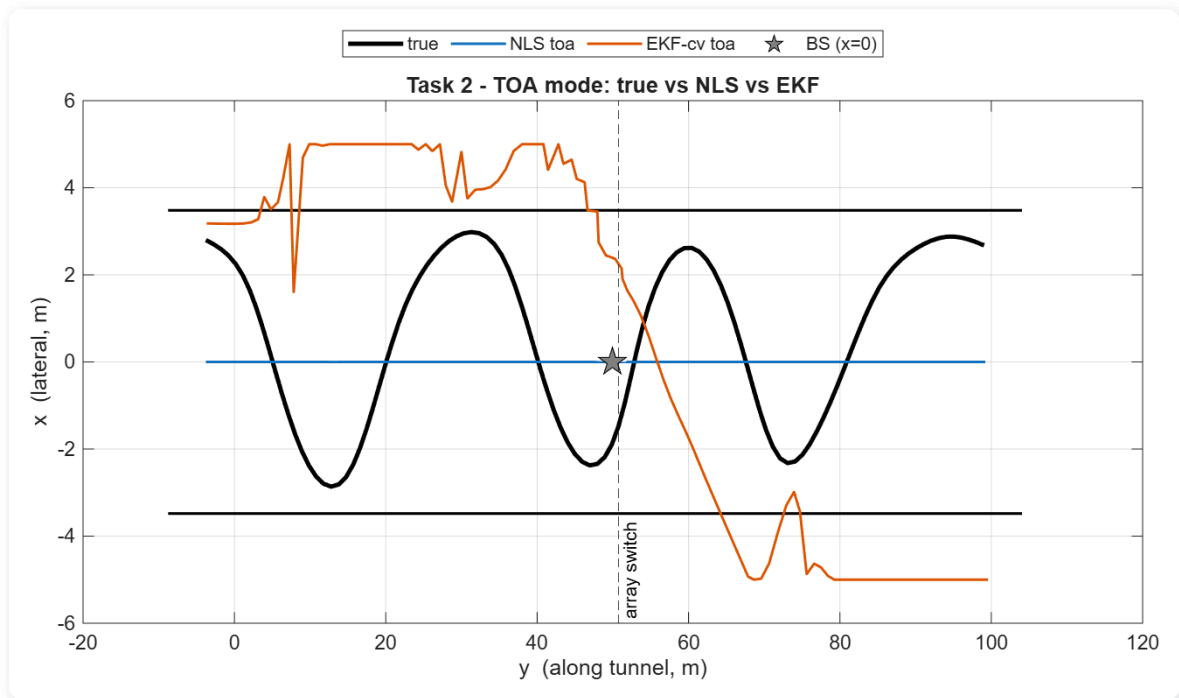
- Constant-velocity model here (motion models compared in Task 3).
- States (x, y) with $z = 1.5$ m known; R from Task 1; χ^2 gating per step.

mode	estimator	RMSE (m)	median	max
ToA	NLS	3.61	2.08	31.9
ToA	EKF-CV	4.96	4.56	7.9
AoA	NLS	4.91	0.42	25.6
AoA	EKF-CV	0.81	0.62	2.0
Joint	NLS	2.84	0.17	29.0
Joint	EKF-CV	0.40	0.25	1.2

Same measurements, 7× lower RMSE: filtering carries information across steps; snapshot NLS cannot.



Why the modes differ: observability at a single BS



ToA-only: NLS pins $x = 0$ (unobservable); the EKF's lateral velocity drifts to the tunnel bounds

EKF-CV ToA per-axis RMSE

0.46 **4.94**

y (m) — range axis x (m) — blind axis

EKF-CV Joint per-axis RMSE

0.12 **0.38**

y (m)

x (m)

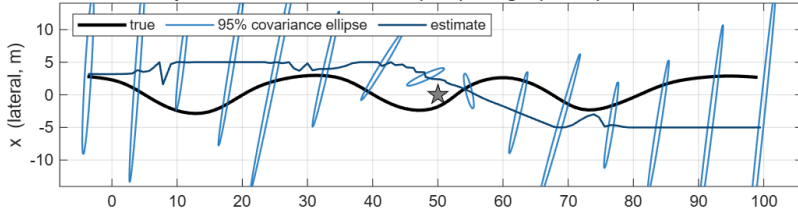
ToA fixes y, AoA fixes x —
complementary by construction.



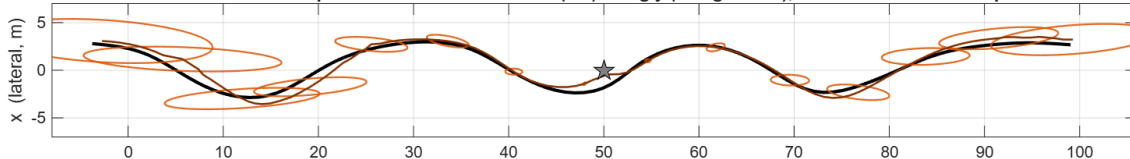
The filter knows what it cannot see: covariance ellipses

Task 4 - what the filter knows it knows: EKF-CV position covariance (no truth used to draw the ellipses)

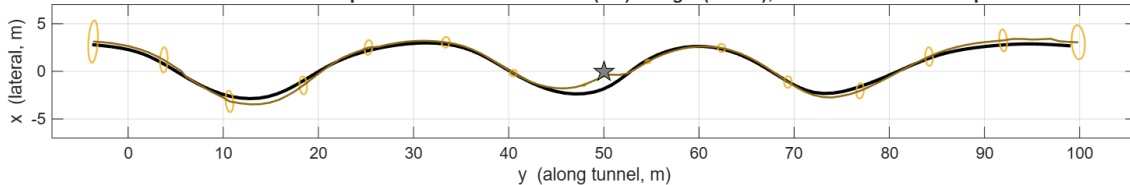
TOA — 95% ellipse semi-axes 9.6 m x 0.4 m (26:1) along x (lateral); truth inside 24% of steps



AOA — 95% ellipse semi-axes 3.6 m x 0.7 m (5:1) along y (along tunnel); truth inside 84% of steps



JOINT — 95% ellipse semi-axes 0.8 m x 0.4 m (2:1) along x (lateral); truth inside 88% of steps



Ellipse shape = observability

ToA 26:1 sliver, sideways

AoA 5:1, along tunnel

Joint 2:1, small + round

Truth inside the 95% ellipse

88%

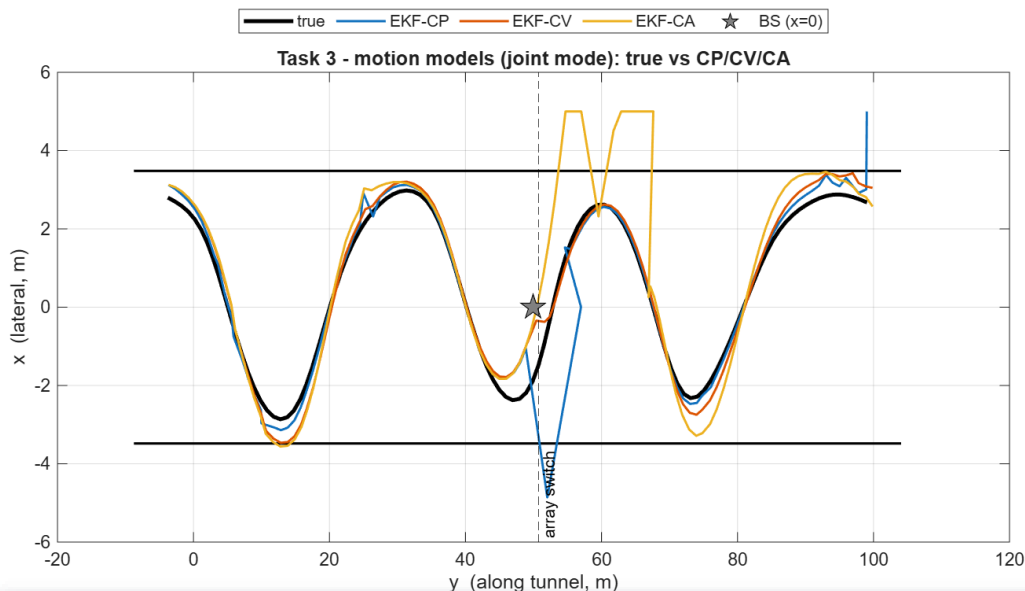
joint — calibrated

24%

ToA — inconsistent



Task 3 — Motion models: CP vs CV vs CA



0.90

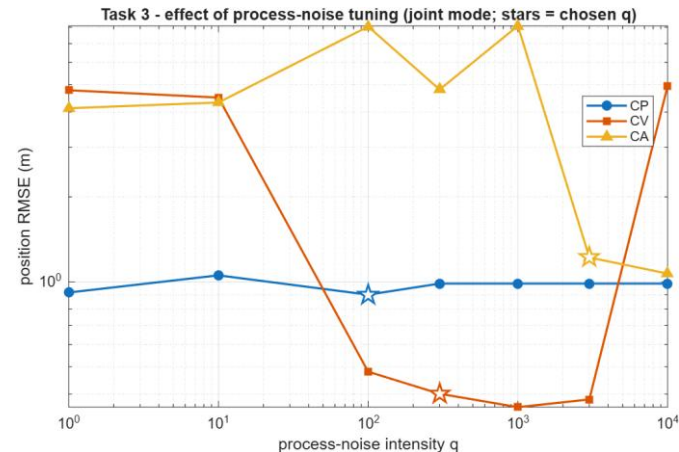
CP · joint RMSE (m)

0.40

CV · joint RMSE (m)

1.22

CA · joint RMSE (m)

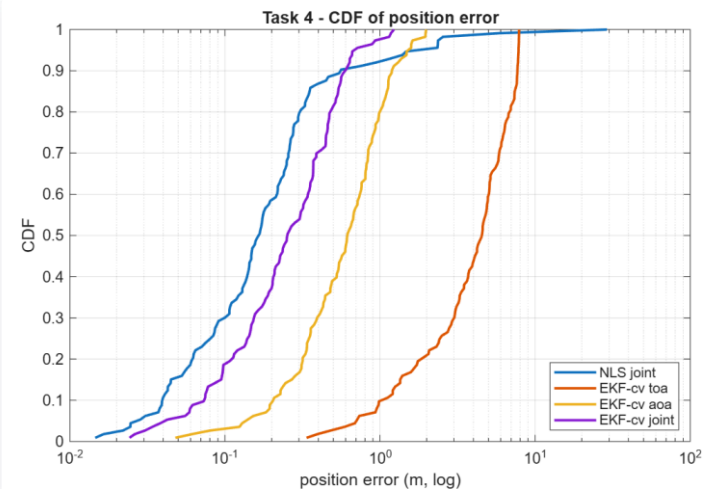
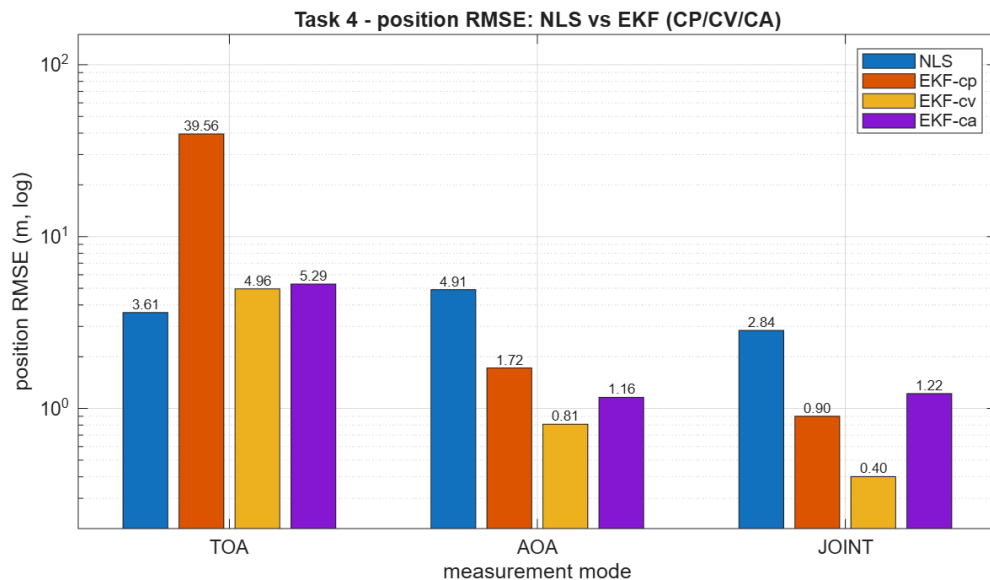


q -sweep: broad CV basin $\Rightarrow$ robust tuning ($\pm 3\times$ barely moves RMSE)

CV matches the physics. CP has no velocity memory — it lags and spikes at the 1-ray BS pass. CA's extra states integrate noise near the singularity (19 gated steps).



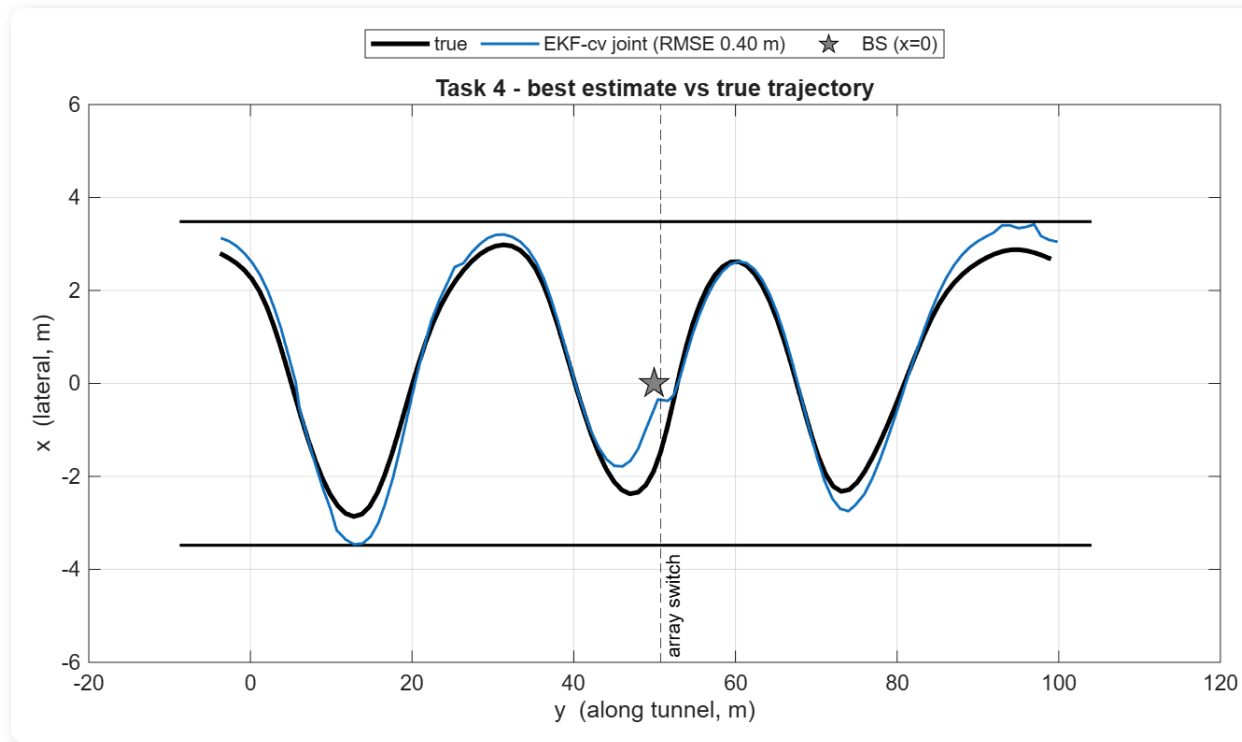
Task 4 — Evaluation: all 12 configurations



Three ordered findings: (1) joint > single sensor for every estimator; (2) EKF $\gg$ NLS — the CDF shows NLS-joint actually has the best median (0.17 m) but a 30 m outlier tail, while EKF-CV-joint is nearly vertical: errors from 3 cm to 1.2 m, predictably small; (3) CV > CP > CA among motion models.



Best real-time result: EKF-CV joint — 0.40 m RMSE



Sub-lane-width accuracy at 98 km/h — 0.4% of the 110 m track

0.40 m

RMSE

0.68 m

95th percentile

1.23 m

worst step



Live animation: [results/video](#)

Conclusions

- ✔ **ToA and AoA are complementary** — range fixes the along-tunnel axis, angles fix the lateral one — joint fusion wins in every estimator (best: EKF-CV joint at 0.40 m RMSE).
- ✔ **Filtering beats snapshot estimation** — same measurements, 7× lower RMSE and a 25× smaller worst case; χ^2 gating absorbs the $t = 19$ ghost ray that breaks every NLS variant.
- ✔ **Match the motion model to the physics** — $CV > CP > CA$: velocity memory carries the filter through the 1-ray BS pass; extra CA states only integrate noise.
- ✔ **Unobservability corrupts the confidence too** — the ToA-only filter's covariance ellipse contains the truth just 24% of the time — a drift is a bias, and a Kalman P cannot represent it.
- ✔ **Results are verified, not just good** — model reproduction vs truth, exact Jacobians, NIS + ellipse-coverage consistency, determinism, robustness to init and $\pm 3\times$ tuning.

Thank you — questions welcome